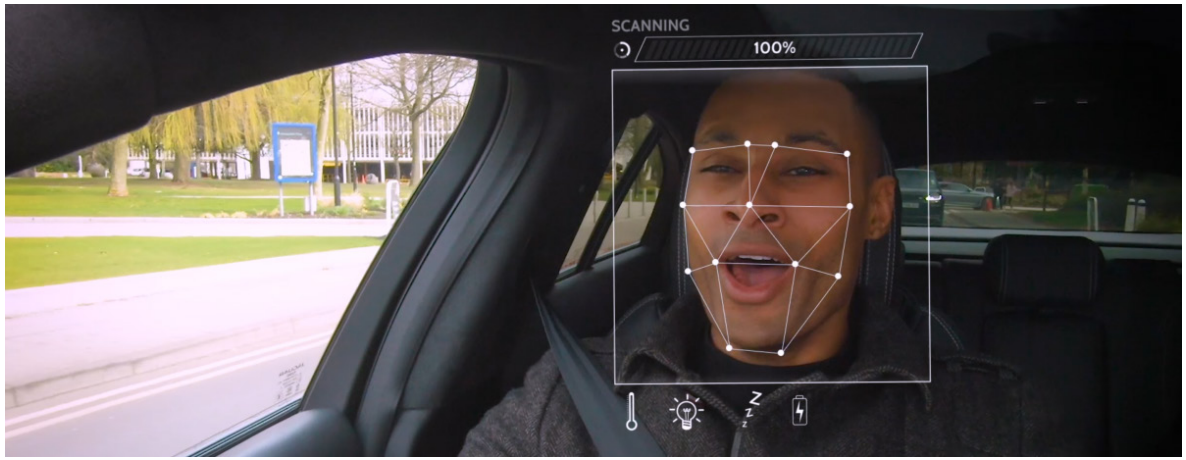
1. The Car That Responds to Your Mood (2018-2019)



"Jaguar Land Rover is researching new artificial intelligence (AI) technology to understand our state of mind while driving – and adjust cabin settings to improve driver wellbeing.

The technology uses a driver-facing camera and biometric sensing to monitor and evaluate the driver's mood and adapt a host of cabin features, including the heating, ventilation and air conditioning system, media and ambient lighting. The settings will be altered in response to the driver's facial expressions to help tackle stress. [...]

The mood-detection system will use the latest AI techniques to continually adapt to nuances in the driver's facial expressions and implement appropriate settings automatically. In time the system will learn a driver's preference and make increasingly tailored adjustments."

This project is an ongoing project that I work on at Jaguar Land Rover.

Source: https://www.jaguarlandrover.com/news/2019/07/car-responds-your-mood-new-jaguar-land-rover-tech-helps-reduce-stress

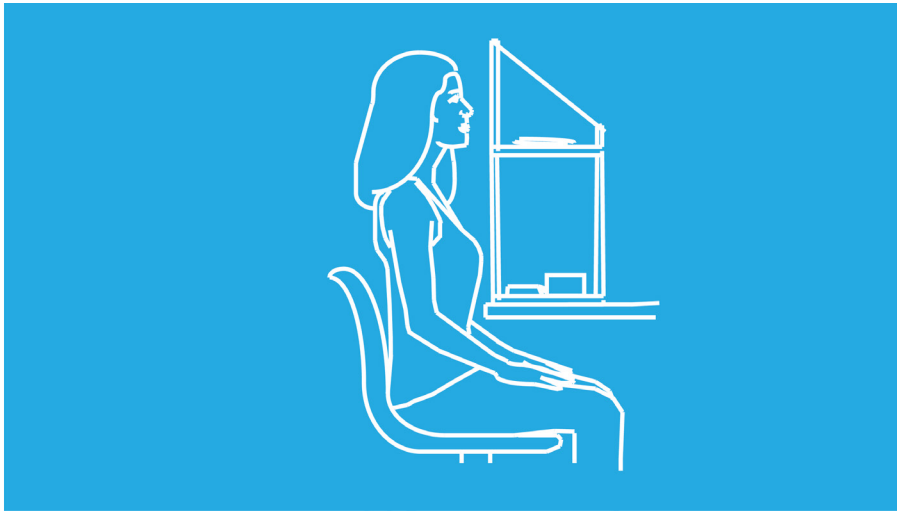
2. Perception of Perspective in Augmented Reality Head-Up Displays (2018)

In this project, I performed two perception experiments to understand depth perception around the perspective cue, in a two-dimensional Head-Up Display.

The outcomes of the experiments are currently being finalized in the form of an academic publication.

The image shows a schematic depiction of one of the experimental setups.

This work was part of my master's thesis in Artificial Intelligence, which resulted in a grade of 8.6/10. The project was done in collaboration with the University of Cambridge. The work will be published in fall 2019 (paper currently under embargo - draft available).



3. 3D Augmented Reality Head Up Display (2017-2019)

This project contained several sub-projects with the end-goal to realise a 3D Augmented Reality full-windscreen Head Up Display.

The project won several company-wide innovation awards. It was a global finalist in the Jaguar Land Rover Innovista Awards, and a regional winner and global finalist in the TATA Innovista Awards.

The project is ongoing as part of my role at Jaguar Land Rover.

Source: https://www.tatainnovista.com

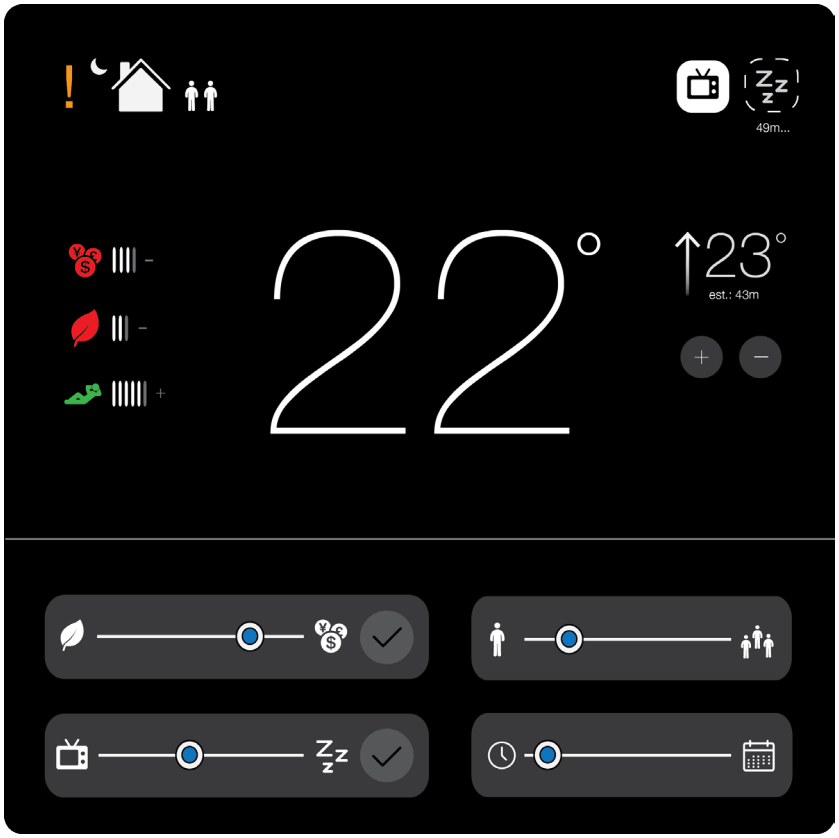
4. (Super) Smart Home UI (2017)

When a smart home setup is developed around a complex artificial intelligence system which learns based on a number of inputs, a new interface design is needed in order to capture what the user's preferences are.

Additionally, the interface needs to provide a clear understanding of the system status and learned preferences, in order to prevent the system to be perceived as an opaque black box.

A graphical user interface was designed to include additional preference controls and information regarding system status of a complex smart home system. The design process gave insights to the technical side of the system, and the design served as a design probe during semi-structured qualitative interviews, which were used to inform engineering decisions.

This project was done during an internship at the NTU IoX Center in Taipei, summer 2017, and was graded 7.5/10 by Utrecht University as part of my M.Sc. in AI.



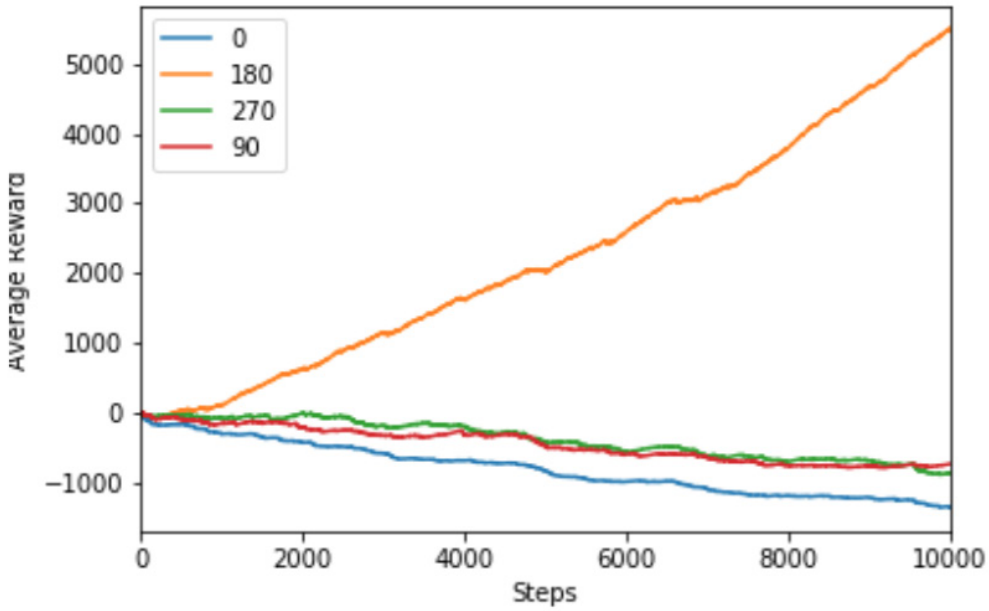
5. Reinforcement Learning Collision Model (2017)

The assignment was to model a situation of skaters on a torus rink, that at each timestep move in a certain direction. When the skaters come too close to each other, collision may happen. Using reinforcement learning, the skaters learn which direction leads to the lowest chance of colliding.

The developed model was written in Python with the Mesa library and executed in a Jupyter notebook. Implemented was an epsilon-greedy Q-learning algorithm. The skaters could choose out of four directions to move in: 0 degrees, 90 degrees, 180 degrees or 270 degrees.

The model shows that in this scenario the skaters almost always converge to an angle of 180 degrees. This result is logical: when an agent turns 180 degrees every time it moves, the least space is involved (as opposed to going in circles or straight ahead) and thus the risk of collision is lowest.

This project was part of an AI course in Multi-Agent Learning and done in pairs during my M.Sc. program. Grade 7.3/10. Full code: https://github.com/bremers/SkaterCollide



Graph above: Average cumulative reward (across agents) for each of four strategies, plotted against the number of steps (i.e.: learning iterations). Over time, the cumulative reward for the strategy of the 180 degree direction becomes significantly higher than for the other strategies, based on which the agents will have learned to favor picking the 180 degree direction strategy.

6. Turing Test for Emotional Intelligence (2016)

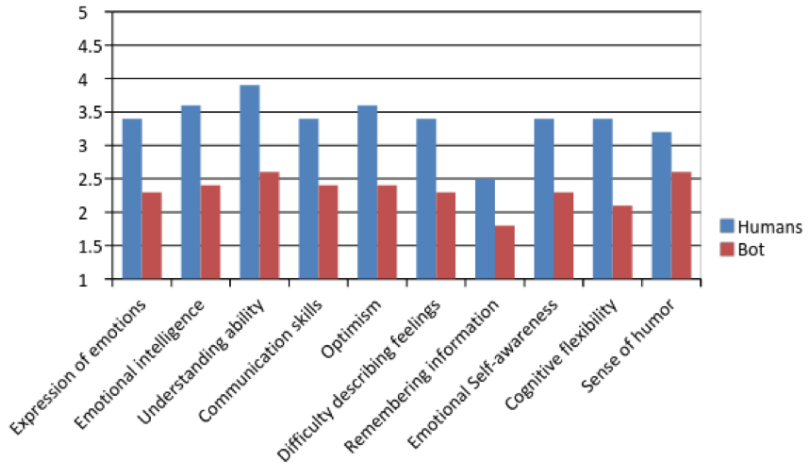
The Turing Test is one of the most well-known tests to judge whether a computer has intelligence. The original design of the test is simple: a participant holds two chat conversations, one with a human and one with a computer. The participant is asked to judge which of the two is human. If the participant thinks that the computer is human, the computer has passed the test.

This test has flaws. For instance, the computer can trick a person into believing it is human by use of tricks that are not necessarily intelligent. Also, the participant can be biased in his way of questioning the other two because he knows that one of the two is a computer.

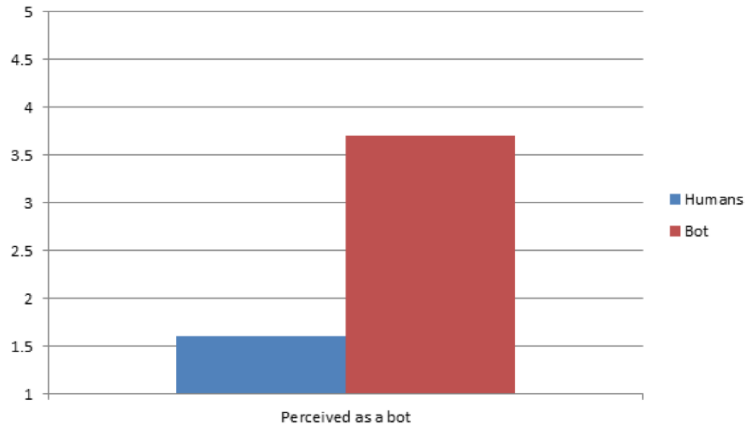
To work around this, we designed a new Turing Test where the participant has simultaneous chat conversations with 3 others (2 human, 1 computer) and is instructed to judge the others on emotional intelligence. At the very end of the test they were asked if they thought whether one of their conversation partners might be a computer. The used chatbot turned out to have passed the test in some cases.

This was a group project in the first year of my M.Sc. in AI, Utrecht University, and was graded 8.4/10.

Graph on the right: Average Likert Scale score of the judged aspects of emotional intelligence displayed by humans and by a bot (1=very low, 5=very high),



Graph on the right: Average Likert Scale score of the extent to which the conversation partner was judged to likely be a bot (1=very kow, 5=very high).



7. How to embroider a radio (2016)

The project brief was to explore textile craftsmanship in relation to digital production methods. I developed a method to make AM crystal radio receivers based on machine-embroidered coils.

When a person covers the receiving coil with one hand and the antenna (also being an embroidered coil) of a transmitter with the other, he could listen to music. Out of this I created pillows that could receive music without the need for a power source.

The coils were created based on a Processing program that generated a .pdf file based on a specified number of windings and diameter of the coil, which was then converted into .dst and embroidered with a digital embroidery machine and multi-kernel isolated copper thread.

This was my B.Sc. graduation project at Industrial Design TU/e, Wearable Senses lab, and was graded 8/10.

8. A study of Aesthetic Qualities (2015)

This project was part of a course on the expression of aesthetic qualities through visuals and materials. At first, we made a collage around the words ‘fast’, ‘intense’ and ‘repulsive’, based on magazine images and Photoshop. Starting from the collage, the assignment was to develop a set of objects in which the aesthetic qualities from the collage where expressed.

The final objects were made from clay that was polished with a spray painted finish (top right), wood (middle right) and carved potato with alpaca wool (bottom right).

The project was done with a partner during the second year of my B.Sc. in Industrial Design, TU/e.

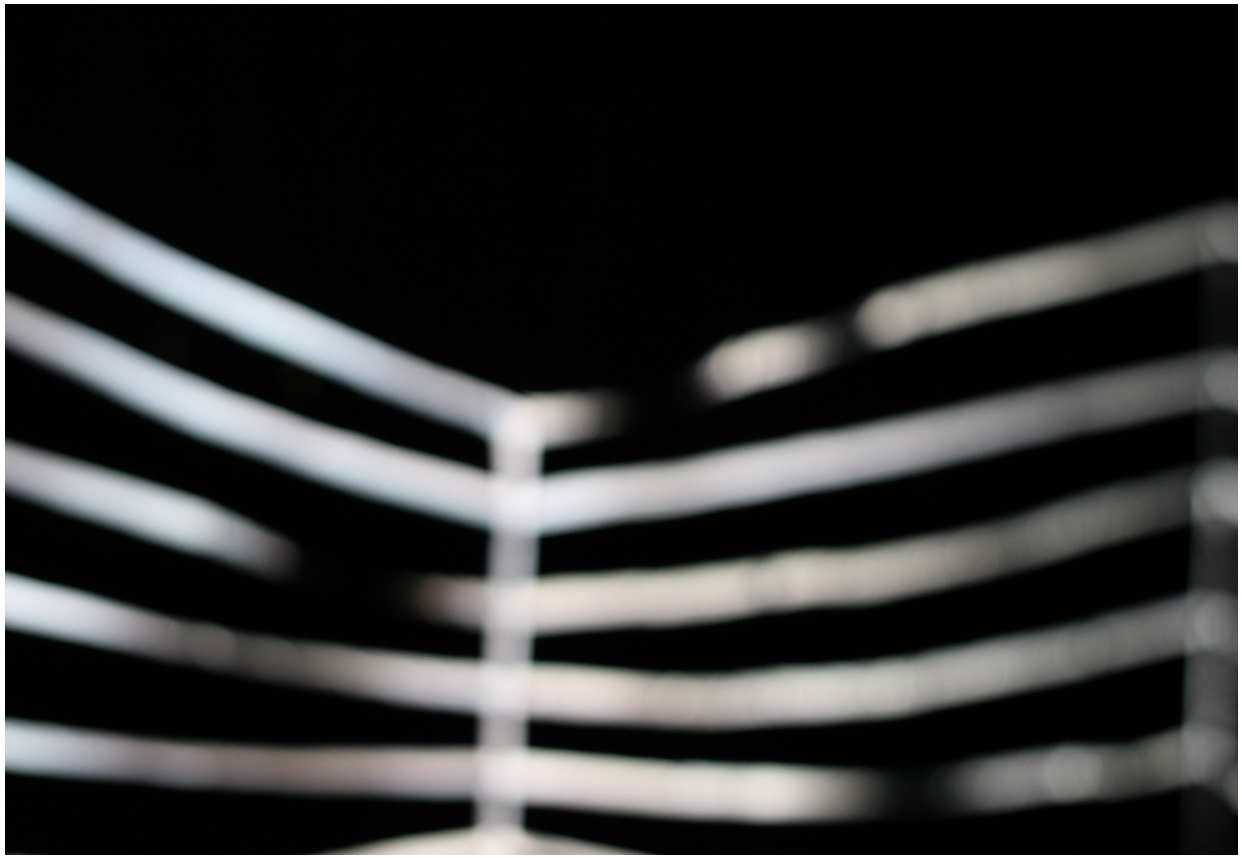
9. Chiaroscuro (2015)

The project brief was to develop a light object or light installation, in which the qualities of natural light or artificial light were shown.

Retro-reflective tape is usually used in traffic and safety for its property to reflect light directly back to the source. This quality is very interesting when the amount of light is extremely low, because then it is only visible from the location of the light source.

I made a darkroom in which orientation was not possible. The walls were covered by a distorted perspective created by retroreflective tape. Multiple visitors would be in the room at the same time, but only one was handed “glasses” (these had no glass and a small LED built-in) who would be able to see the entire work, to the surprise of the other visitors.

I made this in my second year of my B.Sc. in Industrial Design, TU/e, and the project received the mark “Excellent” (top 10%).



10. Internship, Next Nature Network (2015)

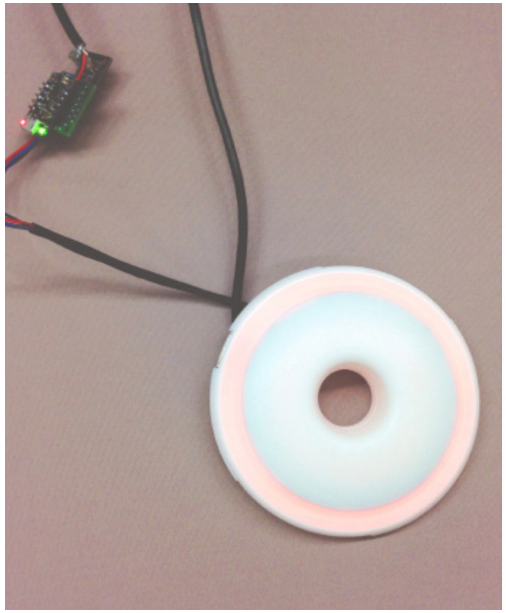
I spent 5 months full time as an intern at Next Nature Network in the fall semester of 2015, which was part of my B.Sc. in Industrial Design at TU/e.

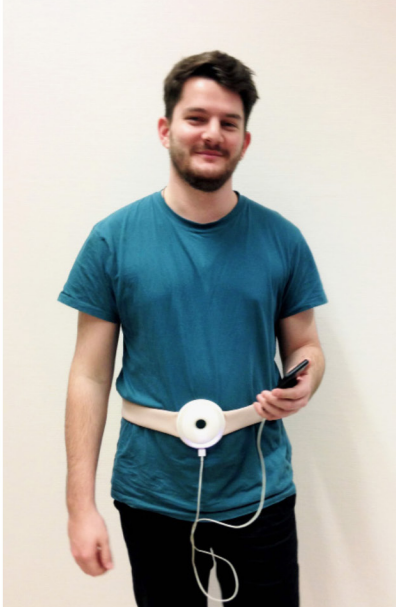
I was responsible for the finalization of the Bistro In Vitro Ice Cart project. The Ice Cart served as a promotion for the Bistro In Vitro book and online project and was exhibited at the Dutch Design Week.

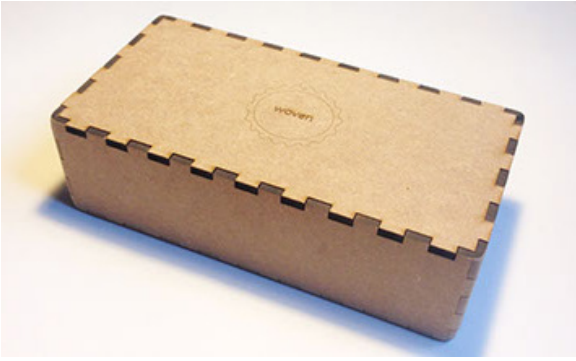
Another project that I worked on was a prototype for the Nano Supermarket exhibition. The prototype was supposed to communicate the story of a hypothetical belt which could be used to charge your phone on belly fat.

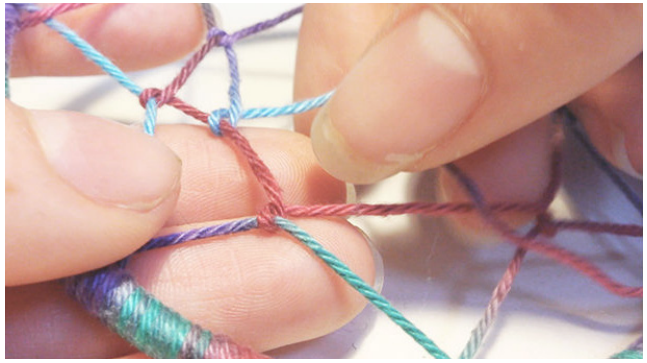
The prototype that we delivered was battery-powered and featured a dynamic light design which could be controlled with a remote for demonstration purposes. I designed this together with another intern.

Apart from these activities I worked on a variety of tasks that arose at the studio.









11. Internetcatchers (2014)

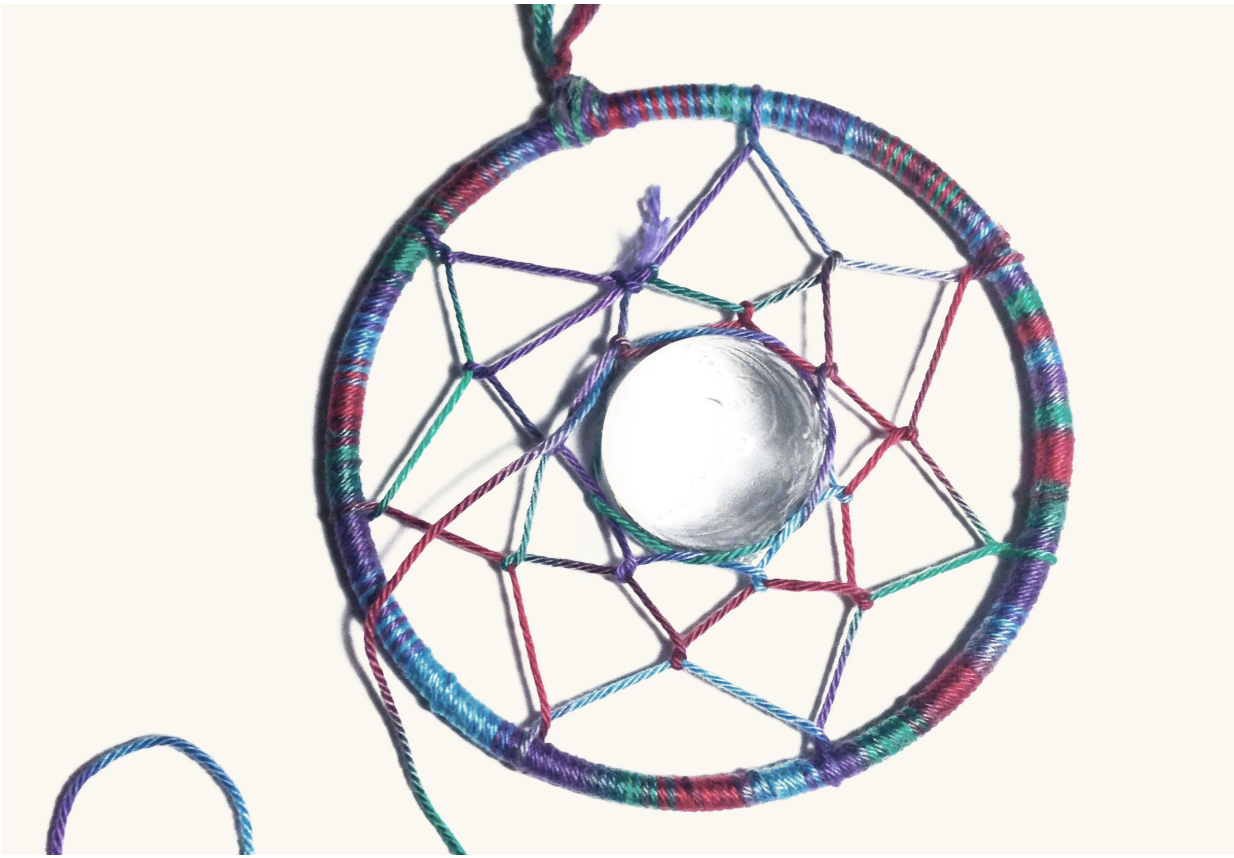
The assignment was to design an artifact that would combine storytelling with novel innovations.

The concept that I arrived at was a toolkit for elementary school children, with which they could weave their own dreamcatchers. These kits would come with a bead that would react with light effects on sensed internet and telecommunication signals.

By letting children realize that ‘virtual’ technologies, such as the internet, have a physical side as well, the first step is made towards winning their interest in the technology and engineering behind it. By letting the children make the object themselves, creativity and pro-activeness (as opposed to with passive forms of entertainment) are encouraged.

I made this in my second year of my B.Sc. in Industrial Design, TU/e, and the project received the mark “Excellent” (top 10%).

A project video can be found at:
https://youtu.be/aEB9kBrhbhY



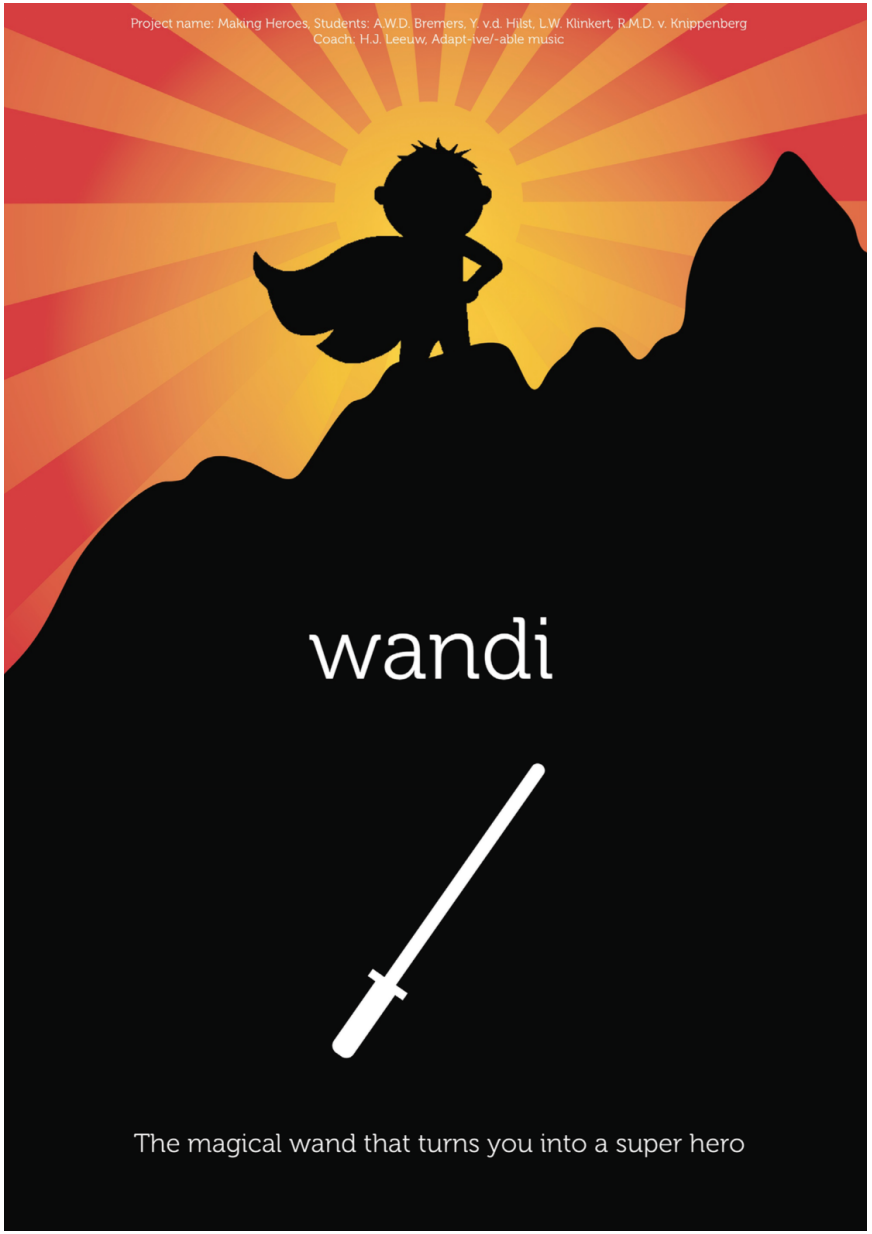
12. Wandi, Kepi (2014)

The aim of this project was to design a musical object for children with disabilities.

From user studies with a patient with Smith-Magenis Syndrome, we learned that among other symptoms, a lot of times these patients suffer from low self-esteem since they realize that they can do less than their normal developing peers. Also, these patients seemed to prefer playing with toys that had sound effects.

We developed the concepts of Kepi and Wandi, a superhero cape and a wand that would accompany movements made by the patient with superhero sound-effects. A technical prototype was made using Teensy, XBees, gyro/acc. sensors and MaxMSP.

This was a team project in my first year of my B.Sc. in Industrial Design, TU/e. A related video that we made during the project can be found at:
https://youtu.be/JkQolk9HODw



13. A Post-Modern Entrance (2013)

As a part of a course on design history, during my B.Sc. in Industrial Design at TU/e, we were assigned to design a new (concept) entrance for the Groninger Museum, based on the ideas of postmodernism.

I started with the idea to use cacti, not only because it would evoke strangeness when placed in front of a museum in Groningen. A cactus is a sturdy plant that survives in difficult climates, that looks dangerous but is crucial since it reveals to man where water is. I found this an interesting metaphor for the role of art in society.

Two cacti could easily serve as pillars for a temple entrance. I added a triangle to resemble the Greek tympanon, as well as two contrasting circle halves to create a more balanced whole.

The result looks comical and ironic, but given that the assignment was on postmodernism, that was more or less the goal.